KK-L-164A
December 8, 1969
SUPERSEDING
Fed. Spec. KK-L-164
June 23, 1964

FEDERAL SPECIFICATION

LEATHER, CATTLEHIDE, PACKING, VEGETABLE TANNED, CHROME TANNED, CHROME-VEGETABLE RETANNED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for full grain and corrected grain cattlehide packing leather.

1.2 Classification.

1.2.1 Tannages, classes and treatments. The leather shall be furnished in the following tannages, classes and treatments as specified (see 6.2).

Tannage I - Vegetable tanned.
Tannage II - Chrome tanned.
  Class 1 - Regular.
  Class 2 - Noncorrosive.
Tannage III - Chrome-vegetable-retanned.
Treatment A - Mildew resistant treated.
Treatment B - Not mildew resistant treated.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

VV-G-109 - Gasoline, Unleaded.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 311 - Leather, Methods of Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal specifications required by activites outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activites may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

Military Specifications:

MIL-L-43432 - Leather, Packaging and Packing of.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

ASTM D-471 - Change in Properties of Elastomeric Vulcanizates Resulting from Immersion in Liquids.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Material. The leather shall be made from cattlehides.

3.2 Tannage.

3.2.1 Tannage I. The leather shall be vegetable tanned.

3.2.2 Tannage II. The leather shall be chrome tanned.

3.2.3 Tannage III. The leather shall be tanned with chrome and vegetable tanning materials.

3.3 Impregnation. When specified (see 6.2), the leather shall be impregnated with waxes, resins or other compounds applicable to conditions of intended use (see 6.1 and 6.3).

3.4 Physical requirements.

3.4.1 Trim (form and area). The leather shall be furnished in the form of bends or belting butt bends and shall conform to the definition of these terms as specified under "Definition of Terms" in Fed. Std. No. 311. The area expressed in terms of square feet shall be legibly marked in the butt area on the flesh side of the leather.

3.4.2 Weight and thickness. The weight and thickness of the leather shall be as specified (see 6.2) and as shown in table I. The leather may be shaved or split to provide greater uniformity of thickness.

TABLE I. Weight and thickness

Weight	Thickness (ounces)
Light	4 to 5-3/4
Medium	6 to 9-3/4
Heavy	10 to 12-3/4
Extra heavy	13 to 16-3/4

NOTE: 1 ounce = 1/64 inch.

3.4.3 Finish. The leather shall be full grain or corrected grain cattlehide. Corrected grain leather shall be buffed or snuffed only to the extent necessary to remove superficial blemishes. The flesh side shall be smooth and free of loose flesh. The leather shall not be embossed or printed.

3.4.4 Breaking strength. Not less than 80 percent of the specimens tested shall meet the minimum breaking force values shown in table II when tested as specified in 4.3.

TABLE II. Breaking strength

Thickness (ounces)	Breaking force (pounds) minimum		
	Tannage I	Tannage II	Tannage III
4 to 5-3/4	65	80	60
6 to 9-3/4	90	110	85
10 to 12-3/4	120	140	110
13 to 16-3/4	150	170	135

3.4.5 Cracking. The leather shall not crack when tested as specified in 4.3.

3.4.6 Shrinkage temperature (tannages I, II and III). When tested as specified in 4.3, tannage I leather shall have a minimum shrinkage temperature of 75°C and tannages II and III leathers shall have minimum shrinkage temperatures of 99°C.

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3.4.7 Area stability (tannages II and III). The area shrinkage of the leather specimens shall be no greater than the percentage values shown in table III when tested under the conditions outlined in table III and as specified in 4.3.

TABLE III. Area stability

Medium	Temperature	Time	Area shrinkage (percent) maximum
Gasoline	23 ± 2°C	60 min.	5.0
Oil	100 ± 1°C	2 min.	10.0
Air	95 ± 1°C	60 min.	5.0
Water	Boiling water	5 min.	5.0

3.5 Chemical requirements.

3.5.1 The leather shall conform to the chemical requirements of table IV when tested as specified in 4.3.

TABLE IV. Chemical requirements

Characteristic	Tannage I		Tannage II				Tannage III	
			Class 1		Class 2			
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Total ash (percent) 1/	-	2.0	-	-	-	-	-	-
Chloroform-soluble materials (percent) 1/ 3/	12.0	20.0	3.0	10.0	3.0	10.0	7.0	15.0
Water-soluble materials (percent) 1/	-	12.0	-	-	-	-	-	-
Degree of trannage 1/	-	65.0	-	-	-	-	-	-
pH value	3.3	-	3.3	-	3.3	-	3.3	-
Chromic oxide (percent) 2/	-	-	4.5	-	4.5	-	3.3	-
Total ash, percent of chromic oxide 1/	-	-	-	130	-	120	-	130

1/ On moisture-free basis.
2/ On moisture-free and chloroform-soluble free basis.
3/ This requirements is not applicable when the leather is impregnated (see 6.2 and 6.3).

3.5.2 Treatment A. When treatment A is specified (see 6.2), the leather shall contain not less than 0.18 percent nor more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.3.

3.5.3 Corrosive effect of leather (tannage II, class 2). The leather shall be noncorrosive when tested as specified in 4.3.

3.6 Workmanship. The leather shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with Fed. Std. No. 311, except where otherwise indicated hereinafter.

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4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. The classification of defects found during the examination of the leather for visual and dimensional characteristics shall be in accordance with 4.2.2.1 and 4.2.2.2. The sample unit for these examinations shall be one bend or belting butt bend. The lot size shall be expressed in units of one bend or belting butt bend. Sampling for these examinations shall be performed in accordance with Fed. Std. No. 311 except that a sample size of 15 and a rejection number of 2 shall be applicable for thickness determinations on lots consisting of 91 or more units of product. Separate examinations shall be performed for visual and dimensional characteristics, but the same sample may be used for each examination.

4.2.2.1 Examination for visual characteristics.

Examine	Defect
Material	Not as specified.
Trims	Not bends or belting butt bends.
Finish	Grain of leather not discernible.
	Embossed or printed leather.
	Flesh side has loose flesh.
	Flesh side not smooth.
Quality of leather	Any hole, deep cut, open scratch, brand, hip mark, wrinkle or other imperfection that indivdually or collectively cause more than a 5 percent loss in the total area of the leather.

NOTE: Minor cuts, soft scars, light well healed scratches and grub holes, light fat wrinkles and slight stains shall not be classified as defects.

4.2.2.2 Examination for dimensional characteristics.

Examine	Defect
Area (expressed in terms of square feet)	Not as marked on leather.
Marking	Omitted, illegible or not in area specified.
Thickness 1/	Not as specified.

1/ Thickness shall be determined in accordance with method 1011 of Fed. Std. No. 311. No individual thickness measurement shall be less than the minimum or greater than the maximum specified.

4.2.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of MIL-L-43432 to determine that packaging, packing and marking requirements of section 5 of this specification are complied with.

4.3 End item testing. The methods of testing specified in Fed. Std. No. 311, wherever applicable and as listed in table V, shall be followed. Sampling procedure and location from which the sample unit is to be obtained shall be in accordance with Fed. Std. No. 311. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any one of the following conditions exist:

a. More than three test failures occur for either breaking strength or area stability.
b. More than one test failure occurs for the remaining requirements applicable to the sample unit.
c. Any composite test result fails to meet the specified requirement.

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TABLE V. End item testing

Characteristic	Requirement paragraph	Test method	Requirements applicable to Sample unit	Requirements applicable to Composite
Material	3.1	1/	-	-
Tannage	3.2	1/	-	-
Impregnation	3.3	1/	-	-
Breaking force	3.4.4	2021	X	-
Cracking	3.4.5	4.3.1	X	-
Shrinkage temperature	3.4.6	7011	X	-
Area stability	3.4.7	4.3.2	X	-
Total ash	3.5.1	6421 2/ 3/	-	X
Chloroform-soluble materials	3.5.1	6341 3/	-	X
Water-soluble materials	3.5.1	6341 3/	-	X
Degree of tannage	3.5.1	6631 3/	-	X
pH value	3.5.1	6621	-	X
Chromic oxide	3.5.1	6615 4/	-	X
Paranitrophenol	3.5.2	6711	-	X
Corrosive effects of leather (qualitative)	3.5.3	3203 5/	X	-

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.
2/ For tannages II and III the total ash shall be expressed in percent of the chromic oxide present in the leather.
3/ Calculated on the moisture-free basis.
4/ Calculated on the moisture-free and chloroform extractable free basis.
5/ The metal used in testing for corrosion shall be as specified in the end item specification (see 6.2).

4.3.1 Cracking. The leather shall be tested for cracking in accordance with method 4021 of Fed. Std. No. 311 except that the results shall be reported as pass or fail. The diameters of the mandrels used for testing the leather shall be follows:

Thickness of leather (ounces)	Diameter of mandrel (inch)
4 to 5-3/4	1/4
6 to 9-3/4	3/8
10 to 12-3/4	1/2
13 to 16-3/4	5/8

4.3.2 Area stability.

4.3.2.1 Immersion mediums.

a. White gasoline (unleaded) 74 octane grade, conforming to VV-G-109.
b. ASTM oil No. 1 as described in the method of test for change in vulcanizates resulting from immersion in liquids (ASTM Designation: D 471).
c. Distilled water.
d. Air.

4.3.2.2 Test specimen. The specimen shall be 51 ± 2 millimeter die cut square of leather taken from the sample unit. Four specimens shall be taken from each sample unit. Only one specimen from each sample unit shall be tested in the same type medium. Specimens to be tested in a liquid medium shall have a small hole not in excess of 0.32 centimeter in diameter punched on the centerline of the specimen, 0.64 centimeter from the edge. The specimens shall be conditioned in accordance with atmospheric conditions for testing as specified in Fed. Std. No. 311.

a. Place the specimen to be measured, grain side up, on a smooth surface of contrasting color, and place a ruler divided into 1.0 millimeter intervals on the grain side applying just enough pressure to flatten the specimen. Measure the specimen to the nearest 0.5 millimeter on the grain side along each of the four sides at a distance of 0.3 ± 0.1 centimeter from the edge and report the average value. Compute the area by squaring the average value for the sides.

b. Pass a copper wire through the hole in the specimens to be tested in a liquid medium. Not more than six specimens shall be placed on the wire. Specimens to be tested in water or oil shall be presoaked by suspending them in 800 ± 10 milliliters of the liquid medium for a period of 16 to 24 hours at room temperature. Specimens to be tested in gasoline shall not be presoaked. The specimens shall be not closer than 2.0 centimeters to the surface of the beaker to which the heat is applied. Space the specimens with brass nuts of 0.4 to 0.5 centimeter thickness. Also prevent the specimen from touching the wall of the beaker by means of spacers.

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c. Heat the fluid (except gasoline) at the rate of 3° to 5°C (5.5° to 9°F) per minute until it reaches the test temperature and keep it at the test temperature for the length of time specified in table III. Testing in an air medium shall be made by laying the specimens flat on a wire screen and placing them immediately into a draft oven at the maximum temperature specified. Use power stirrer for oil. When water is used, stir with a glass rod until the boiling temperature is reached. When gasoline is used, the test shall be made in a hood at 23° ± 2°C away from flames or other sources of ignition.

d. At the end if the test time, remove the specimens from the fluid or oven. Wipe the specimens to remove the excess fluid, and allow them to air dry under atmospheric conditions for testing (see Fed. Std. No. 311) until moisture equilibrium is reached. Compute the area as described in "a" of this paragraph and record.

e. Calculation:

Calculate the percentage area change as follows:

$$\text{Area change, percent} = \frac{A - B}{A} \times 100$$

Where: A = initial area.
B = final area.

f. Report.
The area change shall be reported to the nearest 0.1 percent.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Leather shall be packaged in accordance with the applicable requirements of MIL-L-43432.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. Leather shall be packed in accordance with the applicable requirements of MIL-L-43432.

5.3 Marking.

5.3.1 Civil requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-L-43432.

6. NOTES

6.1 Intended use. When impregnated with a compound applicable to conditions of intended use. the leather may be used for packing reciprocating and rotation parts of hydraulic machinery. When operating temperatures exceed 50° Centigrade, vegetable tanned leather should not be specified. It is believed that tannage II, class 1, leather would cause no appreciable corrosion in ordinary installations. Tannage II, class 2 leather should be specified only for delicate mechanisms or when operating conditions warrant its use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number and date of this specification.
(b) Tannage, class and treatment (see 1.2.1).
(c) When impregnation is required (see 3.3).
(d) Weight and thickness required (see 3.4.2).
(e) Metal to be used in testing class 2 leather for corrosion (see table V).
(f) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Impregnation. In most cases it will be necessary to add waxes, resins or other materials to the leather before it is suitable for use as packing. This will be required when specified in the invitation for bids. Under these conditions the requirement for chloroform-soluble materials is not applicable.

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION Leather, Cattlehide, Packing, Vegetable Tanned, Chrome Tanned, Chrome-Vegetable Retanned - KK-L-164A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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